

# Supplementary Materials

## Complementary Information Channels for Synthetic Lethality Prediction

**Table S1. Top-20 Predicted SL Pairs — Multi-Source Evidence Annotation**

Table 1: Top-20 predicted SL pairs checked against SynLethDB (known SL), NER co-mention, and DepMap CRISPR co-dependency

| Rank | Gene A | Gene B  | Score | SynLethDB | NER | DepMap |
|------|--------|---------|-------|-----------|-----|--------|
| 1    | H3C13  | KDM4D   | 1.000 | —         | —   | —      |
| 2    | H3C13  | ASH1L   | 1.000 | —         | —   | —      |
| 3    | H3C13  | KMT2B   | 1.000 | —         | —   | —      |
| 4    | H3C13  | KDM2A   | 1.000 | —         | —   | —      |
| 5    | H3C13  | SETMAR  | 1.000 | —         | —   | —      |
| 6    | H3C13  | ATF7IP  | 1.000 | —         | —   | —      |
| 7    | H3C13  | RPS6KA5 | 1.000 | —         | —   | —      |
| 8    | H3C13  | KDM5A   | 1.000 | —         | —   | —      |
| 9    | H3C13  | SOX11   | 1.000 | —         | —   | —      |
| 10   | H3C13  | PRMT6   | 1.000 | —         | —   | —      |
| 11   | H3C13  | CDY1B   | 1.000 | —         | —   | —      |
| 12   | H3C13  | ORC1    | 1.000 | —         | —   | —      |
| 13   | H3C13  | BANF1   | 1.000 | —         | —   | —      |
| 14   | H3C13  | TRIM24  | 1.000 | —         | —   | —      |
| 15   | H3C13  | AEBP2   | 1.000 | —         | —   | —      |
| 16   | H3C13  | TRIM33  | 1.000 | —         | —   | —      |
| 17   | H3C13  | PARP1   | 1.000 | —         | —   | —      |
| 18   | H3C13  | ELANE   | 1.000 | —         | —   | —      |
| 19   | H3C13  | BCL2    | 1.000 | —         | —   | —      |
| 20   | H3C13  | SIRT1   | 1.000 | —         | —   | —      |

**Note:** All 20 top predictions involve H3C13 paired with histone-modifying enzymes or chromatin regulators—a biologically coherent module, though none of these pairs appear in SynLethDB, NER co-mention, or DepMap. H3C13 sits at rank 5 in the STRING PPI network by degree (1,358 edges, top 0.03%). This concentration suggests the GNN may over-anchor on highly connected nodes; we discuss this further in the main text.

**Table S2. Negative Sampling Strategy Ablation (Fold1)**

**Note:** PPI-proximal negatives—gene pairs with strong STRING confidence scores but no known SL label—outperform random negatives by 4.2% AUC. In practice, these are harder cases: the

Table 2: Effect of negative sampling strategy on model performance

| Strategy                          | AUC   | $\Delta$ vs Random |
|-----------------------------------|-------|--------------------|
| Random negative sampling          | 0.756 | —                  |
| PPI-proximal structured negatives | 0.797 | +4.2%              |

model must learn that not every pair of interacting proteins is synthetically lethal. Training on them sharpens the decision boundary more than random pairs do.

Table S3. Comparison of 2025 ESM-2-based SL Prediction Methods

Table 3: Concurrent methods that integrate ESM-2 protein language models into SL prediction

| Method | Venue     | ESM-2 Role               | Graph Model          | Cell-line Specific | LLM Component |
|--------|-----------|--------------------------|----------------------|--------------------|---------------|
| ESM4SL | EMBC 2025 | Core en-coder (t33_650M) | Self/Cross-Attention | ✓                  | —             |
| MiT4SL | bioRxiv   | Frozen em-bed+MLP        | PrimeKG HGT          | ✓                  | —             |
| MuSL   | JBHI 2026 | PPI node init            | CNN+GNN+MLP          | —                  | —             |
| Ours   | —         | Frozen embed (t12_35M)   | GATv2 on STRING      | —                  | ✓             |

**Note:** The LLM weak supervision layer is what sets our framework apart—none of the three concurrent ESM-2 methods extract information from the biomedical literature. All three also use larger ESM-2 variants (t33\_650M) than our t12\_35M, so our reported gains from ESM-2 likely represent a lower bound.

Table S4. Ablation — Zero-Vector vs. Shuffled-Feature Comparison

Table 4: Two approaches to estimating text feature importance, compared head-to-head (Fold1)

| Ablation Method                   | AUC (text removed) | Apparent Text Contribution |
|-----------------------------------|--------------------|----------------------------|
| Zero-vector (conventional)        | 0.573              | +22.9%                     |
| Shuffled-feature (fair, proposed) | 0.790              | +1.6%                      |
| Inflation factor                  | —                  | 14.2×                      |

**Note:** Zero-vector ablation overstates text feature importance by more than an order of magnitude. The problem is straightforward: feeding a batch of all-zero vectors into a batch-normalized layer collapses the running statistics, and the resulting instability gets wrongly

chalked up to “the features were important.” Shuffling preserves the marginal distribution of each feature while breaking its association with specific genes, which is a much fairer baseline. We caught this late in the project—earlier drafts of this work used the inflated 22.9% figure—and we recommend checking for this artifact in any GNN with BN layers.

## Table S5. Complete Evidence Levels Across Top-500 Predictions

Table 5: Multi-source evidence broken down by prediction confidence band

| Score Band   | <i>n</i> Pairs | SynLethDB Known | NER Co-mentioned | DepMap Significant |
|--------------|----------------|-----------------|------------------|--------------------|
| [0.5, 0.7)   | 282            | 0 (0%)          | 15 (5.3%)        | 82 (29.1%)         |
| [0.7, 0.9)   | 70             | 0 (0%)          | 6 (8.6%)         | 26 (37.1%)         |
| [0.9, 1.0]   | 148            | 0 (0%)          | 6 (4.1%)         | 39 (26.4%)         |
| <b>Total</b> | <b>500</b>     | <b>0 (0%)</b>   | <b>27 (5.4%)</b> | <b>147 (29.4%)</b> |

**Note:** None of the 500 predictions are in SynLethDB—the model is predicting genuinely new candidate pairs, not rediscovering known ones. About 5% have some literature co-mention, and roughly 30% reach statistical significance in DepMap. The NER-supported pairs cluster in the lower score bands, so the literature signal is providing modest but non-zero orthogonal validation.

## Figure S1. PALB2-XRCC2 in the HR Repair Pathway Network

## Figure S2. NER Co-Mention Network

## Figure S3. PPM1D-TP53 2-Hop PPI Subnetwork

## Appendix A. SLMGAE Head-to-Head Implementation Details

We built SLMGAE as two separate graph encoder branches—one operating on the SL adjacency, one on the PPI adjacency—each with two message-passing layers, followed by an attention gate that learns how much to trust each view. The decoder is a bilinear operation on the fused embeddings. Input features were 256-dimensional random vectors, matching the original paper.

The only thing that differed between our “standard” and “fair” runs was which SL pairs went into the adjacency matrix. For the leaked version, we built the SL graph from all positive pairs before splitting—which is how most published implementations work. For the fair version, we held out every pair where both genes belonged to the test fold.

Everything else—learning rate, optimizer, number of epochs, random seed, PPI graph, training splits—was identical between the two runs. The 13.7% AUC gap between them is purely from the adjacency construction.

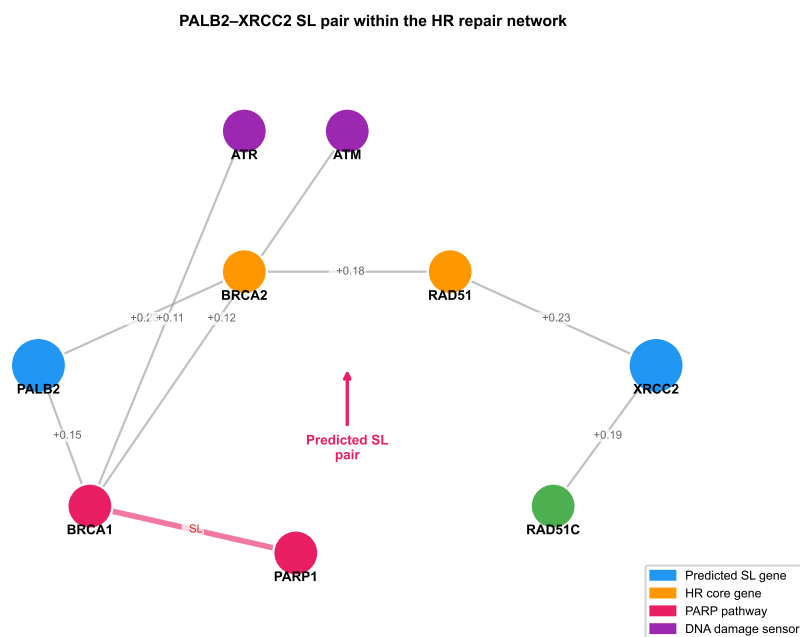


Figure 1: PPI subnetwork around PALB2 and XRCC2 in the homologous recombination repair pathway. Our model predicts these two as co-dependent ( $r=+0.226$  in DepMap,  $p<1e-100$ ). PALB2 bridges BRCA2 to RAD51; XRCC2 functions in the RAD51 paralogue complex. The physical proximity of these two proteins in the HR machinery provides a straightforward structural rationale for their predicted co-essentiality.

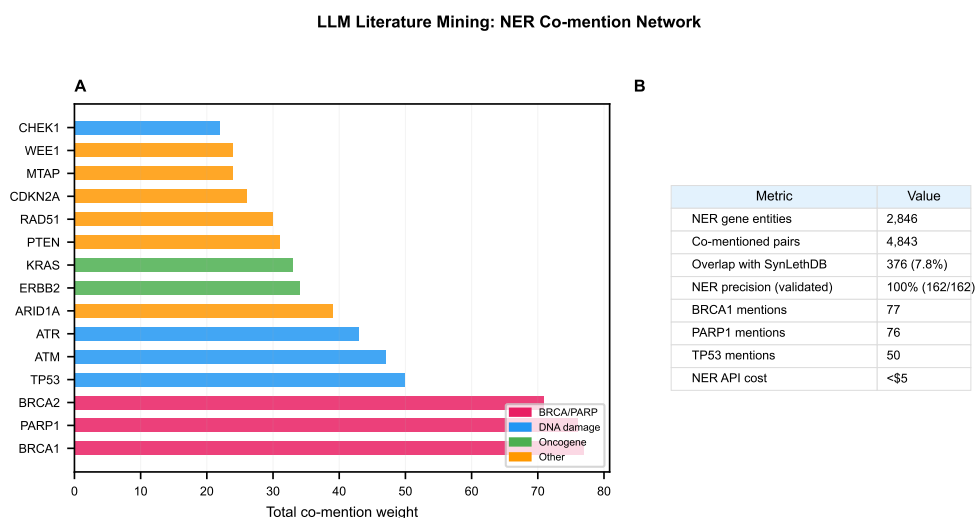


Figure 2: The LLM-derived co-mention network from 1,844 PubMed abstracts. (A) Top hub genes by total mentions: BRCA1 (77), PARP1 (76), TP53 (50). The distribution closely tracks the SL literature—these are exactly the genes one would expect to see heavily discussed in papers about synthetic lethality and PARP inhibitors. (B) Quick stats: 2,846 genes extracted, 2,715 co-mentioned pairs, 7.8% overlap with SynLethDB, 100% precision on the 100-abstract validation set.

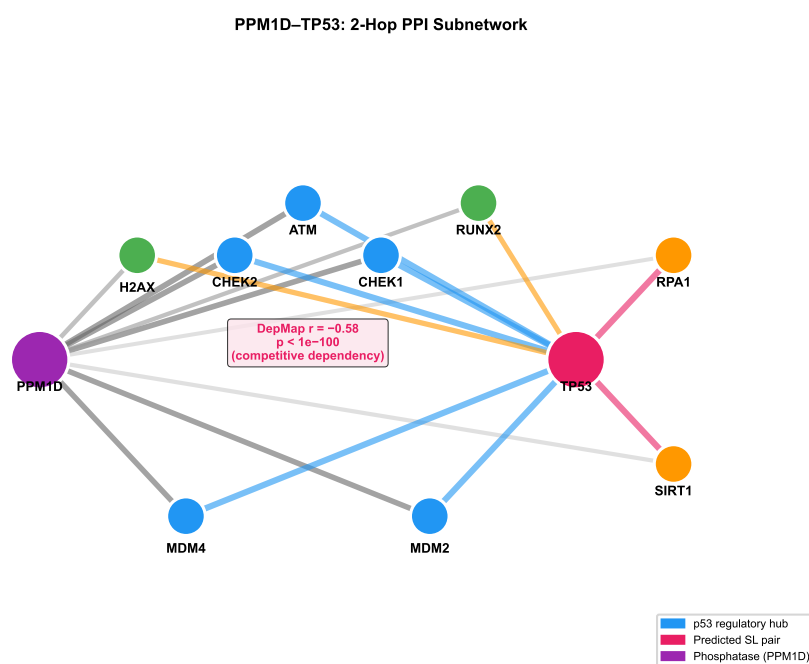


Figure 3: How PPM1D and TP53 connect through the PPI network. Direct neighbors shared by both proteins—MDM2, MDM4, ATM, CHEK1, CHEK2—create seven distinct 2-hop paths between them. These shared interaction partners are all in the p53 signaling and DNA damage response pathways, which is why the GNN assigns this pair a high confidence score even though neither gene’s SL label was in the training data.